

Pedestrians Movement on Crossings

Agent Definition

Definition, characteristics, and interactions of agents in the system

Course: Agent Systems

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1. Agent definition

In the context of this project, an agent represents a single pedestrian navigating through a shared physical environment (a crossing, corridor, or bottleneck). Each agent will be an autonomous entity that perceives its local surroundings, makes independent movement decisions based on behavioral heuristics, and interacts with other agents and obstacles through physical and social forces. There will be no central controller - collective crowd behavior is expected to emerge from individual agent-level rules.

2. Agent characteristics

Each agent will possess the following individual attributes:

- Mass (m) - the physical mass of the pedestrian (planned range: 60-90 kg), used for computing collision forces and body radius ($r = m / 50$).
- Comfortable speed (v_0) - the preferred walking speed (1.5-2.0 m/s) that the agent will try to maintain when unobstructed.
- Vision angle (ϕ) - the field of view (1.2-1.3 rad) within which the agent will scan for obstacles and other pedestrians.
- Maximum vision distance ($d_{\max}$) - the farthest distance (100-105 m) at which the agent will be able to detect obstacles.
- Relaxation time (τ) - the time constant (0.45-0.55 s) governing how quickly the agent adjusts its velocity towards the desired one.
- Desired position - the agent's goal (destination point) on the map.
- Color - a visual group identifier; agents belonging to the same group (i.e., sharing the same origin and destination) will be rendered in the same color.

Each agent will also maintain dynamic state that changes every simulation step: current position, current velocity vector, and desired velocity vector (recomputed by the heuristics at each step). We plan to use a double-buffering mechanism for position and velocity to allow concurrent updates without read-write conflicts.

3. Agent types

All agents will share the same behavioral model - there will be no explicit type hierarchy. However, agents will be differentiated by their parameterization and group assignment:

- Group membership - agents will be spawned in groups defined by their origin area and shared destination. For example, in a planned crossing scenario, four groups could approach an intersection from four cardinal directions, each heading to the opposite side.
- Parameter variation - within a group, agents will be assigned randomized mass, comfortable speed, vision angle, and relaxation time drawn from predefined ranges. This heterogeneity should produce realistic variance in individual behavior.

The architecture is designed to support extending agent types in the future (e.g., a police officer agent with different force parameters, or a child agent with reduced speed and vision range) by varying the initialization parameters.

4. Communication and interactions

4.1 Interaction model

Agents will not communicate through explicit message passing. Instead, all interaction will be implicit and physics-based, mediated through the shared environment. Each agent will perceive the positions and velocities of nearby agents and react according to the Social Force Model rules. This is consistent with real pedestrian behavior, where people do not verbally coordinate but instead react to observed positions and movements of others.

4.2 Types of interactions

Three categories of interactions are planned:

- Agent-to-agent repulsion - when two agents' body radii overlap, a contact force proportional to the overlap distance will push them apart. This prevents physical overlap and models body-to-body collisions in dense crowds.
- Agent-to-wall repulsion - when an agent's body radius intersects with a wall segment, a repulsive force will push the agent away from the wall surface.
- Heuristic-based avoidance - before physical contact occurs, the direction and distance heuristics will cause agents to proactively steer away from obstacles and other pedestrians within their field of view.

4.3 Expected emergent behaviors

Although agents will follow simple local rules, several complex collective behaviors are expected to emerge from their interactions:

- Lane formation - counter-flowing pedestrian streams should spontaneously organize into lanes, reducing collisions.
- Bottleneck congestion - agents should accumulate before narrow passages, forming arching clusters around the opening.
- Freezing by heating - under excessive crowd pressure, mutual blocking should paradoxically reduce flow to near zero.
- Group mixing and separation - at crossing intersections, groups from different directions are expected to temporarily intermix before separating and continuing to their respective destinations.

5. Technical framework

5.1 Programming language and platform

The simulation will be implemented in Java 17+ using JavaFX for the graphical user interface. We plan to use Gradle as the build tool and structure the project as a modular Java application.

5.2 Planned technologies and libraries

- JavaFX - GUI framework for rendering the simulation canvas, drawing agents as colored circles, walls as lines, and providing configuration controls.
- Java Concurrency (ExecutorService, thread pools) - we plan to support both single-threaded and multi-threaded computing engines for parallel agent updates.
- JUnit 5 - intended for unit testing of core model classes and utility functions.

5.3 Data

The simulation will not rely on external datasets. All input data will be generated procedurally through map initializers that define wall layouts and agent spawn configurations. We plan to create several predefined maps, ranging from

simple single-direction flow scenarios to complex multi-way crossing intersections. Agent parameters (mass, speed, vision) will be drawn from configurable uniform distributions at initialization time.